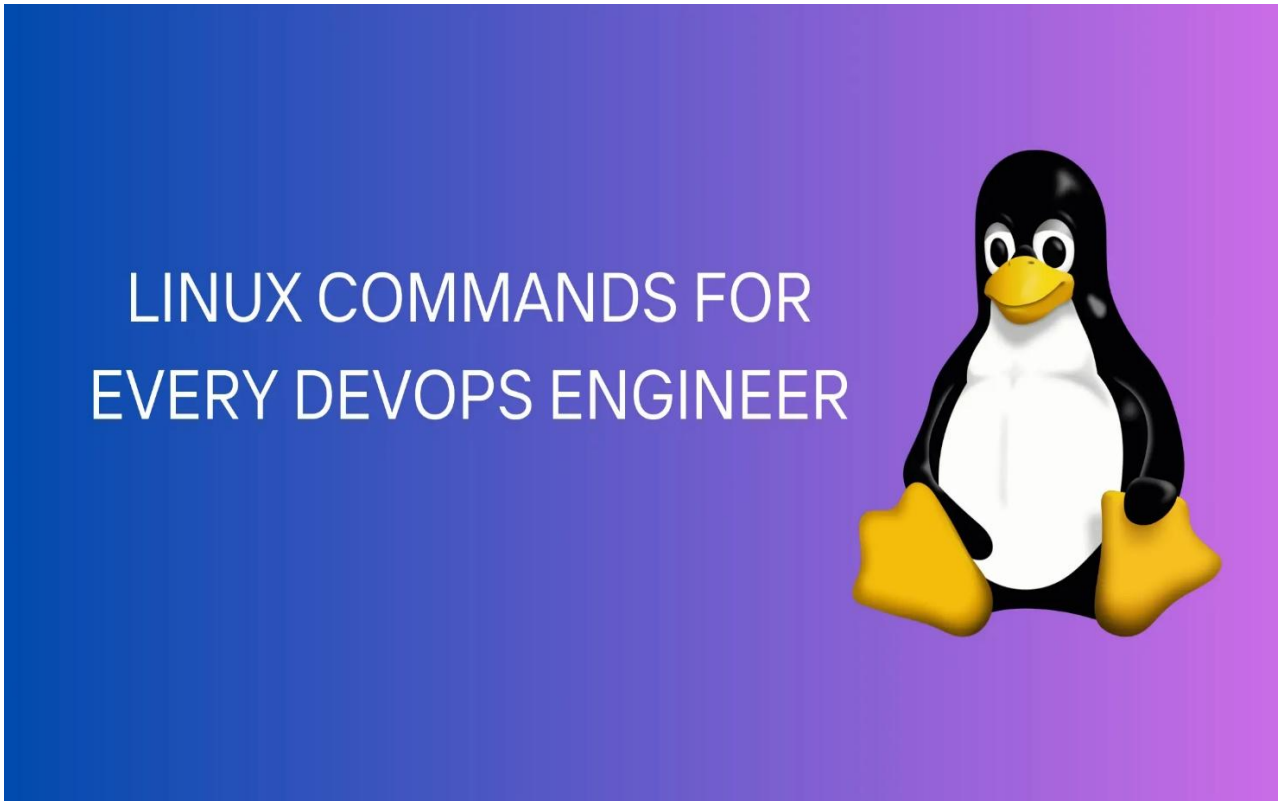


Introduction to Linux

Linux is a free and open-source operating system known for its high security, stability, and reliability.
It is a multi-user operating system, which means multiple users can access and work on the same system simultaneously without affecting each other's work.

Linux is widely used in:

- Servers
- Cloud platforms
- DevOps environments
- Cybersecurity
- Embedded systems
- Enterprise infrastructure

Operating System (OS):

An Operating System (OS) is system software that acts as an interface between computer hardware and the user.

Functions of an OS:

- Provides a platform for applications to run
- Manages CPU, memory, storage, and devices
- Controls processes and system resources
- Enables user interaction with hardware

Applications such as browsers, MS Office, Notepad, games, and software tools require an operating system environment to execute.

Without an operating system, a user cannot communicate with the computer hardware.

Kernel:

The **Kernel** is the core (heart) of the operating system.

Responsibilities:

- CPU management
- Memory management
- Device management
- Storage management
- Process management

It is the **lowest level** of the operating system and directly interacts with the hardware.

Daemons:

Daemons are background services that run automatically in the system.

Examples:

- Printing services
- Sound services
- Scheduling services
- Network services

They start during system boot or after user login and continuously run in the background.

Shell:

A **Shell** is an interface that allows users to interact with the operating system.

Functions:

- Accepts user input
- Executes commands
- Runs programs and scripts
- Displays output

Types of Shells:

- **CLI (Command Line Interface)** – Text-based commands
- **GUI (Graphical User Interface)** – Windows, icons, buttons

The shell acts as a bridge between the **user and the operating system**.

Command:

A **command** is an instruction given to the operating system to perform a specific task.

Example:
ls, cd, pwd, mkdir, rm

Terminal

A **Terminal** (CLI / Console) is a text-based interface that allows users to:

- Enter commands
- Execute operations
- Receive output
- Interact with Linux systems

It exists in Linux, macOS, and Windows (Command Prompt / PowerShell).

Linux Distributions

Linux distributions are modified versions of Linux created for different use cases.

Popular Distributions:

- RedHat
- Ubuntu
- Debian
- CentOS
- Fedora
- OpenSUSE
- Kali Linux
- Amazon Linux
- Rocky Linux

Linux History

- **1991** – Linus Torvalds (University of Helsinki, Finland) started developing Linux
- Initially named **Freax**, later renamed to **Linux**
- Written in **C language**
- Released under **GNU General Public License (GPL)** in 1992

Today Linux is used in:

- Supercomputers
- Smartphones
- Servers
- Cloud platforms
- Laptops
- Tablets
- Smart TVs
- Cars
- Routers
- Home appliances

Open Source Concept

Linux follows the **Open Source philosophy**:

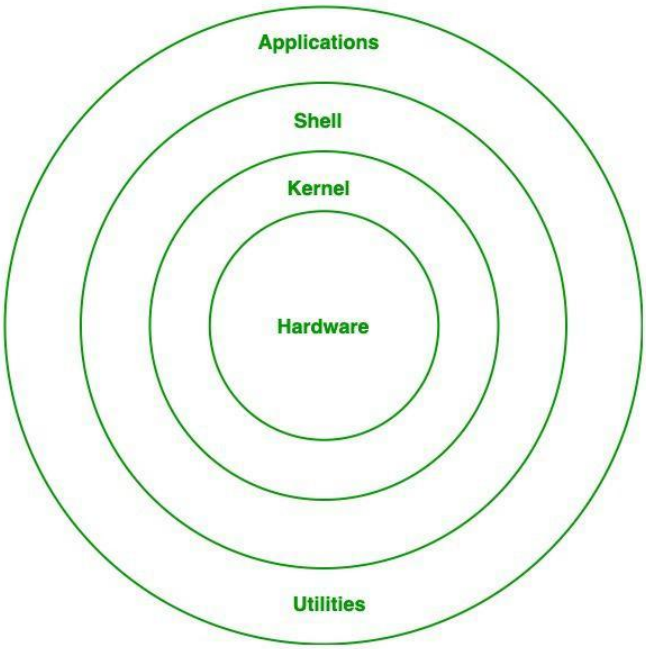
Freedoms:

- Freedom to use
- Freedom to study
- Freedom to modify
- Freedom to distribute
- Freedom to share modified versions

Linux Architecture

Linux consists of:

1. Hardware
2. Kernel
3. Utilities
4. Shell
5. Applications



Practical Linux Usage

Root User Access

Root is the superuser with full permissions.

sudo -i

Exit root:

exit

Command Categories:

- System Commands
- Hardware Commands
- File Commands
- Folder Commands
- Permission Commands
- User Commands
- Group Commands
- Search Commands
- Networking Commands

System Commands:

uname

uname -r

uname -a

clear

uptime

uptime -p

hostname

hostname -i

hostnamectl set-hostname "swiggy"

ip addr

ip route

ifconfig

date

timedatectl

timedatectl set-timezone Asia/Kolkata

who

whoami

ps

kill -9 PID

Date Formats:

date +"%d"

date +"%m"

date +"%y"

date +"%H"

date +"%M"

date +"%S"

date +"%D"

date +"%F"

date +"%A"

date +"%B"

Hardware Commands:

lscpu

cat /proc/cpuinfo

lsblk -a

free

cat /proc/meminfo

free -m

df -h

File Commands:

touch filename

touch aws azure gcp

touch linux{1..5}

rm filename

rm -f filename

rm -f aws azure gcp

rm -f linux{1..5}

rm -f *

rm -f *.txt

rm -f a*

Folder Commands:

mkdir foldername

mkdir git maven jenkins

mkdir docker{1..5}

rmdir foldername

rmdir git maven jenkins

rmdir docker{1..5}

rmdir *

rm -rf *

Listing Files:

ll

ls

ll -t

ll -r

ll -a

Directory Navigation:

cd foldername

cd

cd -

cd ../

cd ../../

Directory Structure:

mkdir folder1/folder2

ll folder1

touch folder1/aws.txt

mkdir -p aws/azure/gcp/devops

Copy Commands:

cp source destination

cat file1 >> file2

Move / Rename:

mv file1 file2

Cat Commands:

cat filename

cat > filename

cat >> filename

Editors

Vim Editor

Modes:

1. Command Mode
2. Insert Mode
3. Save & Quit Mode

Command Mode:

gg, G, M, 4gg, 17gg, :23

:set number

yy, 4yy

p, 10p

dd, 5dd

u, ctrl+r

/word, ?word

:%s/old/new/

Insert Mode:

i, A, I, O, o

Save/Quit Mode:

:w

:q

:q!

:wq

:wq!

Grep Command:

grep "word" filename

grep -n "word" filename

grep -c "word" filename

grep -i "word" filename

User Commands:

cat /etc/passwd

getent passwd

useradd username

userdel username

userdel -r username

useradd -M username

su - username

passwd username

Group Commands

cat /etc/group

groupadd groupname

groupdel groupname

usermod -a -G group user

Permission Commands:

chown username filename

chgrp groupname filename

chown username:groupname filename

chown -R username:groupname folder

chmod 777 filename

Search Commands:

Find:

find . -name file

find /proc -name filename

find . -type d -name folder

find . -type f -name file.txt

find . -type f -perm 777

find . -type f ! -perm 777

find / -user username

find / -group groupname

Locate

locate filename

Conclusion

Linux is the foundation of modern IT infrastructure and the backbone of the **DevOps ecosystem**.
It powers cloud platforms, containers, automation tools, CI/CD pipelines, and enterprise systems.

Mastering Linux enables:

- Automation
- Scalability
- Reliability
- Security
- Performance